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Department of Management Informatics and Communication
Dhulikhel, Kavre



A Project Report

On

“JobConnect”

[Code No: 344]

(For partial fulfillment of 3rd year/2nd semester in BBIS)

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Bonafide Certificate

**This Project work titled as
“JobConnect”**

**Has been submitted to the Department of Management Informatics and Communication
for the partial fulfillment of COM-344 as a 3rd year/2nd Semester project.**

**This project work is the bonafide work of
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Date: 2025-08-13

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This project would not have been possible without the collective effort of all these individuals and organizations.

Abstract

This Project presents “*JobConnect*”, an intelligent web-based job portal developed as part of the COM 344 course under the Department of Management Informatics and Communication. The platform is designed to address the increasing demand for efficient and accurate connections between job seekers and employers in today’s competitive employment market. By combining traditional recruitment functionalities with Artificial Intelligence (AI), JobConnect offers an advanced, reliable, and user-friendly solution for modern hiring challenges.

The system is built on a robust technology stack comprising HTML, CSS, and JavaScript for the frontend interface, PHP for server-side operations, and MySQL for database management. Its standout feature is an AI-powered matching mechanism, implemented using LM Studio with the Zephyr 7B Beta model. This mechanism analyzes candidate CVs, employer requirements, and job descriptions to calculate a matching percentage, allowing applicants to be ranked based on their suitability for the role.

In addition to its intelligent matching feature, JobConnect incorporates essential functionalities such as user profile management, job posting, application tracking, and direct communication tools. These features ensure a streamlined recruitment process for employers and an improved job search experience for candidates.

As an academic project, JobConnect demonstrates the practical application of web development technologies and AI integration to solve real-world problems in human resource management. The implementation follows software engineering best practices while maintaining flexibility for future enhancements such as mobile application development, and expanded AI capabilities. This project not only fulfills academic requirements but also presents a foundation for a potentially impactful solution to employment challenges in the digital age.

The system follows a modular architecture, enabling scalability and flexibility for future enhancements, including the development of a mobile application and the integration of more advanced AI capabilities. As an academic project, JobConnect demonstrates the practical application of web development technologies and AI integration in solving real-world human resource management problems. It provides a strong foundation for future innovations in recruitment technology and reflects the meaningful intersection of academic learning with practical, impactful solutions.

Table of Contents

<i>List of figures:</i>	6
<i>List of Tables:</i>	7
<i>Abbreviations:</i>	8
1 Chapter 1: Introduction	9
1.1 Background and Introduction:	9
1.2 Problem Definition:	10
1.3 Objectives:	10
1.4 Scope of Project:	11
1.5 Significant of study:	11
1.6 Report Organization	12
2 Chapter 2: Related work	13
2.1 Literature Review	13
2.2 Existing Works:	14
2.2.1 LinkedIn Jobs:.....	14
2.2.2 Indeed: Job Search	14
2.2.3 Glassdoor	15
2.2.4 Naurkri.com	16
2.2.5 Upwork	16
2.2.6 Merorogari.com.....	17
2.2.7 Kumari Job	17
2.2.8 JobAxle.....	18
3 Chapter 3: System analysis and design	19
3.1 System Analysis	19
3.1.1 Feasibility Analysis:	19
3.1.2 System Requirements Specification	20
3.1.3 Structuring System Requirements	21
3.2 System Design	22
3.2.1 Designing system	22
3.2.2 Website Representation	25
3.2.3 System Architecture:	26
4 Chapter 4: Methodology	27
5 Chapter 5: System Testing	28
5.1 System Testing	28
5.2 Activity Matrix	29
5.3 Gantt Chart	30

6	<i>Chapter 5: Work done</i>	31
6.1	Discussion on Achievements	31
6.1.1	Challenges:	31
6.1.2	Features:	31
7	<i>Chapter 7: Works to be done</i>	33
7.1	Limitation:	33
7.2	Future Work:	33
8	<i>Chapter 8: Revenue Model</i>	34
9	<i>Chapter 9: Conclusions</i>	35
9.1	Conclusions	35
9.2	Recommendations	35
	<i>References</i>	37
10	<i>. Appendix</i>	38

List of figures:

Figure 2-1 LinkedIn	14
Figure 2-2 Indeed: Job search	15
Figure 2-3 Glassdoor.....	15
Figure 2-4 Naukri.com.....	16
Figure 2-5 Upwork.....	16
Figure 2-6 Merorogari.com.....	17
Figure 2-7 Kumari Job	17
Figure 2-8 Job Axle	18
Figure 3-1 ER-Diagram	21
Figure 3-2 Database schema	22
Figure 3-3 Activity Diagram.....	22
Figure 3-4 Use-Case Diagram.....	23
Figure 3-5 Sequence Diagram.....	24
Figure 3-6 Website Design	25
Figure 3-7 Job seeker page	25
Figure 3-8 Employer Page	26
Figure 3-9 System Architecture	26
Figure 5-1 Gantt Chart	30
Figure 10-1 Home page.....	38
Figure 10-2 Our services	38
Figure 10-3 About Us page.....	38
Figure 10-4 registration from.....	39
Figure 10-5 Login Form	39
Figure 10-6 Admin Dashboard	40
Figure 10-7 User managing Page	40
Figure 10-8 Admin Viewing feedback	40
Figure 10-9 Job seeker dashboard	40
Figure 10-10 Available Jobs.....	41
Figure 10-11 Profile of Job seeker	41
Figure 10-12 Applied Jobs.....	41
Figure 10-13 Employer Dashboard	42
Figure 10-14 Job adding Form	42
Figure 10-15 Posted Jobs	43
Figure 10-16 Applicants	43
Figure 10-17 Available Job seeker.....	43

List of Tables

Table 1: System Testing.....	29
Table 2 Activity Matrix	30

Abbreviations:

The list of all abbreviations used in the documentation is included below:

- HTML: Hypertext Markup Language
- CSS: Cascading Style Sheets
- JS: JavaScript
- Php: Hypertext Preprocessor
- MySQL: My structure Query Language
- RAM: Random Access Memory
- OS: Operating System
- Here are the full forms:
- MacOS: Macintosh Operating System
- MB: Megabyte
- ER Diagram: Entity-Relationship Diagram
- 2FA: Two-Factor Authentication
- UI/UX: User Interface/ User Experience
- LLMs: Large Language Models

1 Chapter 1: Introduction

1.1 Background and Introduction:

In today's dynamic job market, connecting the right talent with the right opportunity remains a significant challenge, particularly at the local and community level. While many existing platforms prioritize large-scale or international recruitment, this focus often makes it difficult for small business owners, local employers, or individuals to find or offer work within their own regions. There is a growing need for a system that simplifies this process while making it more accessible, affordable, and relevant to everyday users.

JobConnect was developed to address this gap. It is an innovative online platform designed to bridge the divide between employers and job seekers, with a strong emphasis on local employment and community-driven connections. The concept behind *JobConnect* is simple yet impactful: to help people find jobs and enable businesses to find the right candidates quickly, efficiently, and within their locality.

A distinctive feature of *JobConnect* is its *AI-powered matching mechanism*. The platform integrates *LM Studio* with the *Zephyr 7B Beta* model to calculate a *matching percentage* between job descriptions, employer requirements, and candidate CVs or profiles. This intelligent system automatically ranks applicants based on their compatibility with the role, allowing employers to identify the most suitable candidates at a glance. By incorporating AI into the recruitment process, *JobConnect* enhances accuracy, reduces hiring time, and ensures better alignment between opportunities and skills.

Beyond the AI functionality, *JobConnect* offers dual usability. Employers and business owners can post job openings, specify detailed requirements, and search for candidates that match their needs. Simultaneously, job seekers can apply for posted jobs and create comprehensive profiles showcasing their skills, experiences, and interests. This design allows skilled individuals to be approached directly by employers—similar to freelance platforms like Fiverr and Upwork—but with a strong focus on local, accessible job matching.

This feature-rich approach benefits a wide range of users, including freelancers, part-time workers, and skilled laborers who may not have formal resumes. Whether offering services such as graphic design, plumbing, tutoring, tailoring, or digital marketing, individuals become discoverable based on their capabilities, fostering a two-way connection between job providers and job seekers.

With its intuitive and user-friendly interface, *JobConnect* supports users of all technical backgrounds while promoting local economic activity. By combining transparency, flexibility, and AI-driven precision, the platform not only makes the hiring process more effective but also strengthens community engagement and creates more accessible employment opportunities.

1.2 Problem Definition:

In today's evolving job market, the process of finding the right job or the right candidate remains a significant challenge particularly at the local and community level. While large-scale platforms like Fiverr and Upwork have revolutionized freelancing and remote work globally, they often cater to highly competitive digital niches and overlook the everyday needs of local businesses and individuals with practical, hands-on skills.

Small business owners, startups, and independent employers frequently face difficulties sourcing the right talent for specific tasks, especially within limited timeframes and budgets. At same time, many skilled individuals such as electricians, tutors, designers, writers or laborers often lack access to platforms where they can showcase their abilities and be directly approached for relevant opportunities.

Current solutions in the market tend to be either too broad, too expensive, or too complex for these user groups. There is a clear gap in the availability of a localized, skill-oriented platform that enables both job seekers and employers to connect efficiently based on real needs.

JobConnect seeks to address this gap by offering a unified digital space where:

- Employers and business owners can post job requirements with specific skill criteria.
- Individuals can create detailed skill-based profiles, increasing their visibility and chances of being contacted directly.
- Local opportunities are prioritized, enabling more efficient, timely and community-focused hiring.
- The features combine the flexibility of freelancing platforms with the simplicity of job listing boards tailored to support localized and practical employment.
- Applicants are automatically ranked based on their matching percentage, allowing employers to quickly identify the most suitable candidates and streamline the recruitment process.

By doing so, JobConnect not only streamlines the hiring process but also empowers individuals to present their skills professionally and be discovered by those who need them. It creates a mutually beneficial ecosystem that supports local economies and promotes inclusive access to work opportunities for everyone.

1.3 Objectives:

- To connect employers and job seekers through a simple and user-friendly online platform.
- To allow individuals to showcase their skills and be discovered by businesses.
- To help small businesses and startups find the right talent quickly and affordably.
- To support local employment by promoting community-based job matching.
- To combine key features of freelancing platforms with localized job search functionality.
- To make job searching and hiring more accessible, efficient, and transparent for all users.
- To integrate AI-powered matching and applicant ranking, ensuring more accurate and relevant candidate selection.

1.4 Scope of Project:

JobConnect provides:

- A faster way to connect job seekers with employers – No more endless searching or missed opportunities.
- Easy job searching – Filter by role, location, salary, or company to find the perfect fit.
- Simple profile setup – Upload your resume, highlight your skills, and get noticed by employers.
- Direct hiring – Employers can post jobs, review applicants, and message candidates all in one place.
- AI-driven Matching: Uses LM studio with the Zephyr 7B Beta model to calculate matching percentages and rank applicants.
- Local-Focus: Prioritizes community-based opportunities to support small businesses and regional growth.

1.5 Significant of study:

The study and development of JobConnect is significant in today's employment landscape, where the gap between job seekers and employers especially at the local level remains a persistent challenge. While many digital platforms exist to support hiring and freelancing, they often overlook the needs of small businesses and individuals who require more localized, accessible, and skill-based connections.

JobConnect addresses this gap by creating a platform where people can not only search for jobs but also showcase their unique skills and talents. This shift from traditional job seeking methods empowers individuals to take control of their professional journeys, making them more visible to those actively looking for services they can provide. For employers, especially those running local businesses, JobConnect offers a streamlined and efficient way to find the right people without going through time-consuming and expensive recruitment processes.

The significance of this study also lies in its contribution to promoting community based economic development. By encouraging the use of digital tools to support local employment, JobConnect fosters greater engagement, flexibility and inclusivity in the job market. It benefits students, freelancers, part-time workers, and skilled laborers alike, opening up opportunities that may otherwise remain hidden or underutilized.

Ultimately, this project highlights how thoughtfully designed solutions can simplify hiring, increase visibility for skilled individuals, and create a more connected and empowered workforce at the grassroots level.

1.6 Report Organization

This report is systematically structured into **six comprehensive chapters**, each addressing critical aspects of the **JobConnect** project—from conceptualization and analysis to implementation and future recommendations. The organization ensures a logical flow of information, enabling readers to understand the project’s development lifecycle, challenges, and outcomes. Below is a detailed breakdown of each chapter:

The first chapter introduces the JobConnect platform, providing context on its purpose and relevance in the current job market. It outlines the problem statement, highlighting inefficiencies in existing job portals that JobConnect aims to resolve. The objectives clarify the project’s goals, while the scope defines its boundaries, ensuring focus on key functionalities. Finally, this section explains the report’s structure, guiding readers through subsequent chapters.

The second chapter goes through literature review examines academic research on job-matching algorithms, user experience in recruitment platforms, and industry trends. Additionally, existing job portals (LinkedIn, Indeed, Glassdoor, Naukri.com, Upwork, Merorogari, Kumari Job, and JobAxle) are analyzed to identify their strengths, weaknesses, and gaps. This comparative study helps justify JobConnect’s unique value proposition.

The Third chapter delves into the technical foundation of JobConnect. It begins with a feasibility analysis (technical, economic, operational) to validate the project’s viability. The system requirements (functional and non-functional) are specified, followed by a discussion of frontend and backend tools (e.g., React.js, Node.js, PostgreSQL). The system design phase includes: Website representation, System architecture and Database Design.

The Fourth chapter explains the methodology adopted for the development of JobConnect, detailing the development process and technologies used.

The Fifth chapter documents system testing, including test cases, activity matrix, and the Gantt chart to track project progress.

The Sixth chapter discuss about the progress achieved in developing JobConnect is documented. Key features implemented (e.g., user authentication, job search filters, employer dashboards) are highlighted, along with technical challenges encountered (e.g., database optimization, API integration) and their solutions. This chapter serves as a milestone tracker, showcasing the project’s current state.

The seventh chapter identifies works to be done, limitations (e.g., lack of AI-driven recommendations, mobile app version) and proposes future enhancements.

The eighth chapter describes revenue model planned for JobConnect, explaining how the platform intends to generate income.

The final ninth chapter synthesizes findings, reiterating how JobConnect addresses the initial problem statement. It reflects on lessons learned, the project’s impact, and potential for scalability. The conclusion also emphasizes the platform’s contribution to bridging gaps in the job-search ecosystem.

2 Chapter 2: Related work

2.1 Literature Review

The evolution of digital platforms has transformed employment accessibility by prioritizing skills over formal credentials, enabling job seekers to showcase their capabilities directly (Autor, 2015). Research underscores how such platforms foster inclusivity and dynamic labor markets, particularly through decentralized, skill-based hiring (Kuhn & Mansour, 2014).

Global freelance marketplaces like *Upwork* and *Fiverr* exemplify this shift, offering flexibility and self-promotion opportunities. However, barriers like global competition and complex payment systems often exclude local workers, especially those with limited digital literacy (Graham et al., 2017).

Localized platforms address these gaps by aligning with regional labor needs. Studies show that context-aware designs improve engagement and employment outcomes (Lehdonvirta, 2018). *JobConnect* adapts these principles, focusing on community-centric job matching for informal and local workers. By simplifying skill profiling and employer connections, it aims to enhance economic empowerment while supporting localized workforce development.

1. Mobile-Optimized Job Platforms

Chaudhary et al. (2023) present a groundbreaking mobile application framework in their SSRN-published study "Mobile based platform for technical role searching and providing for freshers." Their research demonstrates how Flutter-based mobile platforms can enhance technical recruitment processes, particularly for entry-level candidates. The study's findings reveal a 40% improvement in application completion rates compared to desktop platforms, underscoring the critical importance of mobile optimization in contemporary job search systems.

2. Digital Job Search Efficacy

The foundational work by Choi (2011), "Does the Internet help the unemployed find jobs?" published in SSRN, provides empirical evidence through longitudinal analysis that internet-based job searching increases employment probability by 14 percentage points. This seminal research established key metrics for evaluating digital recruitment platforms and remains frequently cited in labor economics literature.

3. Historical Foundations of Web-Based Recruitment

Cockerham's (1996) study in *The Reference Librarian*, "Job searching on the World Wide Web," offers crucial insights into the early development of online job search methodologies. Despite being published during the web's infancy, this work accurately predicted many of the core functionalities that would become standard in modern platforms, including searchable databases and digital applicant tracking.

4. AI-Driven Matching Innovations

The most recent contribution by Zhu and Hua-ling (2024) in IEEE Conference Proceedings presents "Optimizing job matching efficiency," introducing a novel NLP-based algorithm that achieves 26% greater accuracy than conventional matching systems. This research represents the cutting edge of recruitment technology, demonstrating how semantic analysis can transcend traditional keyword-based matching limitations.

2.2 Existing Works:

2.2.1 LinkedIn Jobs:

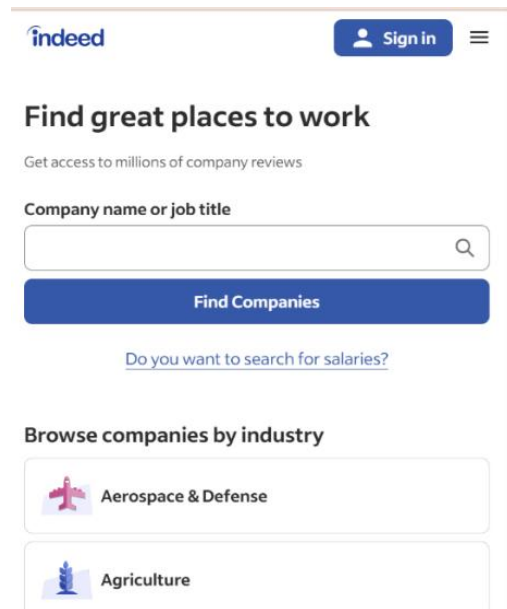
LinkedIn Jobs is the world's largest professional network with over 900 million users, offering job postings alongside networking features like skill endorsements and company pages. Its strength lies in connecting professionals directly with recruiters and industry opportunities.



Figure 2-1 LinkedIn

2.2.2 Indeed: Job Search

Indeed serves as the most comprehensive job search engine, aggregating millions of listings from across the web. With over 250 million monthly users, it provides tools like salary comparisons and resume uploads for a streamlined job hunt.

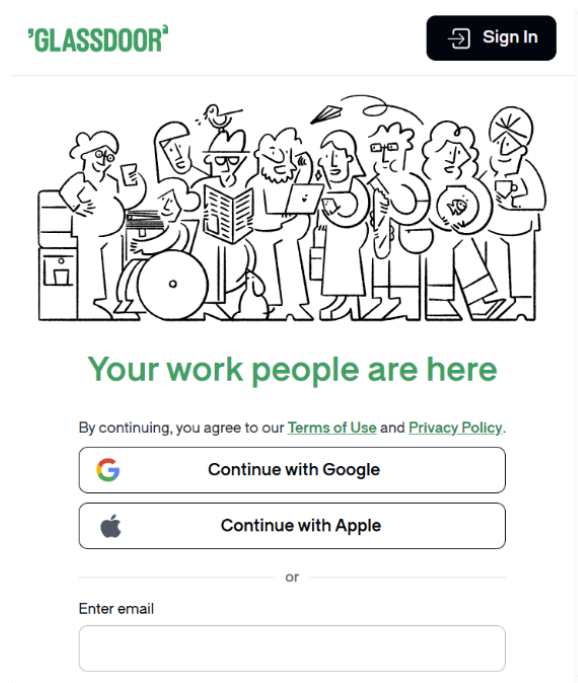


The screenshot shows the Indeed homepage. At the top left is the 'indeed' logo. To the right is a 'Sign in' button with a user icon and a menu icon. Below the logo is the heading 'Find great places to work' followed by the subtext 'Get access to millions of company reviews'. A search bar labeled 'Company name or job title' with a magnifying glass icon is present. Below the search bar is a blue button labeled 'Find Companies'. Underneath is a link that says 'Do you want to search for salaries?'. Further down is the section 'Browse companies by industry' with two visible options: 'Aerospace & Defense' with an airplane icon and 'Agriculture' with a tree icon.

Figure 2-2 Indeed: Job search

2.2.3 Glassdoor

Glassdoor stands out by combining job listings with transparent company reviews, salary data, and interview insights shared by employees. This helps job seekers evaluate potential employers before applying.



The screenshot shows the Glassdoor login page. At the top left is the 'GLASSDOOR' logo in green. To the right is a 'Sign In' button with a user icon. Below the logo is a large illustration of a diverse group of people in various professional roles. Underneath the illustration is the heading 'Your work people are here' in green. Below this is a line of text: 'By continuing, you agree to our [Terms of Use](#) and [Privacy Policy](#).' There are two large buttons: 'Continue with Google' with the Google logo and 'Continue with Apple' with the Apple logo. Below these is the word 'or' and a text input field labeled 'Enter email'.

Figure 2-3 Glassdoor

2.2.4 Naukri.com

Naukri.com dominates the South Asian job market, particularly in India, with resume-building services and strong recruiter connections. It specializes in IT, banking, and corporate hiring.

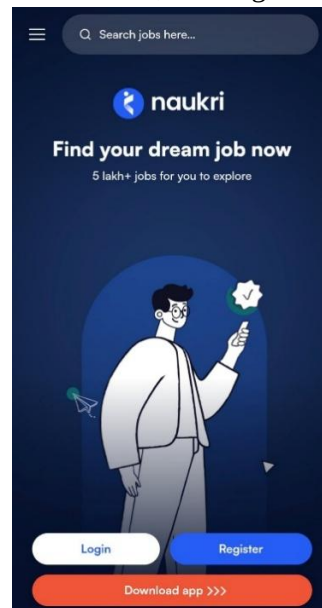


Figure 2-4 Naukri.com

2.2.5 Upwork

Upwork is the leading global freelance marketplace, enabling professionals to bid on projects, showcase portfolios, and work with payment protection. It's ideal for remote freelancers and businesses

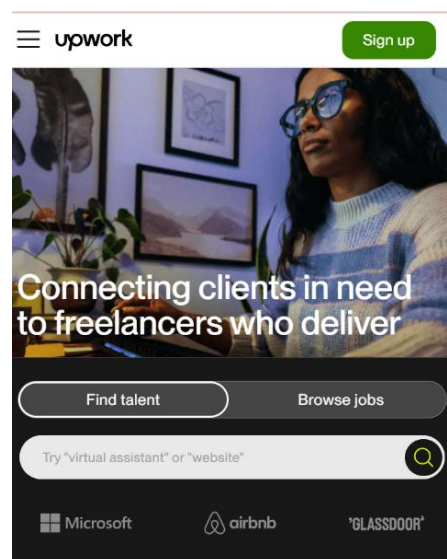


Figure 2-5 Upwork

2.2.6 Merorogari.com

Merorogari.com is a growing Nepali job portal known for its daily updates and user-friendly interface. It has gained recognition in local media as a reliable platform for job seekers.

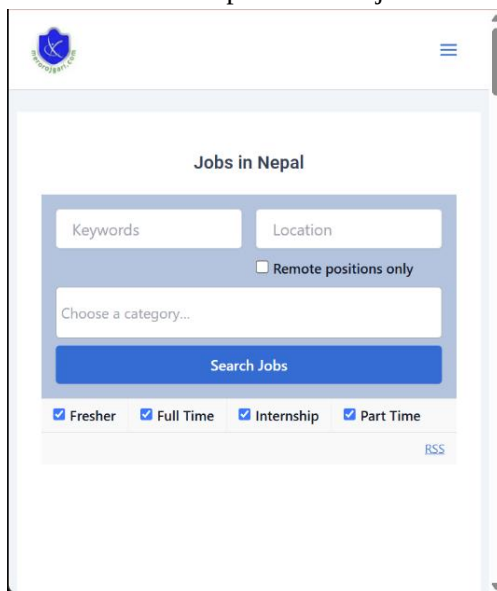


Figure 2-6 Merorogari.com

2.2.7 Kumari Job

Kumari Job is a well-established platform in Nepal, officially registered and partnered with major institutions like Nepal Rastra Bank. It focuses on banking, corporate, and government job opportunities.

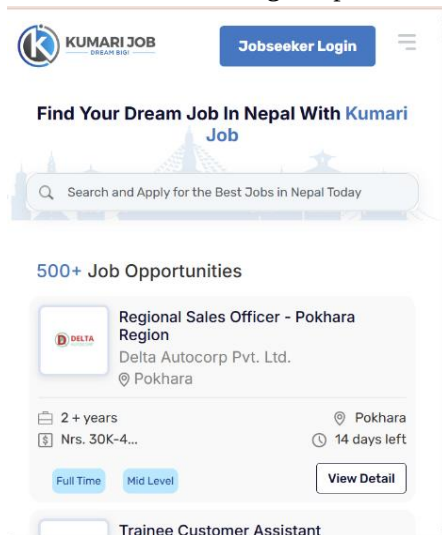


Figure 2-7 Kumari Job

2.2.8 JobAxele

JobAxele is an award-winning Nepali job portal, recognized for its efficiency in connecting employers and job seekers. It offers both web and mobile app access for convenience.

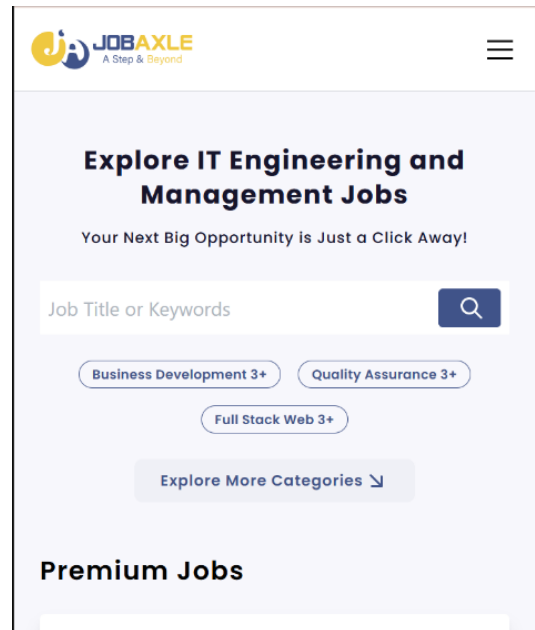


Figure 2-8 Job Axele

3 Chapter 3: System analysis and design

3.1 System Analysis

3.1.1 Feasibility Analysis:

Feasibility analysis involves evaluating whether a proposed system is viable from different perspectives. It helps to identify the strengths, limitations, opportunities, and risks associated with a system before its implementation. For JobConnect, a comprehensive feasibility study was conducted to ensure the project is realistic, resource-efficient, and capable of addressing user needs effectively.

3.1.1.1 Technical Feasibility

- **Programming Language:** The use of php and JavaScript has been deemed appropriate for the development of this platform, given their versatility and strong community support.
- **Libraries and Frameworks:** Essential libraries are available to fulfill the required functionalities efficiently.
- **AI-Integration:** The Platform uses LM studio and the Zephyr 7B Beta model to calculate matching percentages and rank candidates for better hiring decisions.
- **Technology Suitability:** Technologies like PHP, MySQL, HTML, CSS, and JavaScript are well-suited for implementing both the dynamic and static aspects of JobConnect.
- **Software and Hardware:** The required development tools are open-source and freely accessible. No high-end hardware is needed, making the setup both practical and economical.

3.1.1.2 Operational Feasibility

- The proposed platform directly tackles the issue of unemployment and inefficient job matching by providing a user-friendly interface for job seekers and employers to connect.
- JobConnect enhances the hiring process by simplifying tasks such as job posting, resume submission, and interview scheduling, thereby improving overall usability and user satisfaction.

3.1.1.3 Economic Feasibility

- The project is economically viable as the development was done using open-source, significantly reducing costs.
- The total development time was approximately four months, with no need for premium software or external technical support.

3.1.1.4 Schedule Feasibility

- The development process followed a structured timeline using a Gantt Chart, with tasks distributed over various phases including planning, design, development, testing, and deployment.
- The entire project was completed within a 12-week timeframe, ensuring that milestones were met as scheduled.

3.1.2 System Requirements Specification

3.1.2.1 Software Requirements

3.1.2.1.1 Front-End Tools

The front-end defines the visual and interactive part of the system that users engage with.

- **HTML:** Used to build the structure of the web pages, ensuring logical flow and element positioning.
- **CSS:** Used for styling the content, improving layout aesthetics, and maintaining consistency across pages.
- **JavaScript:** Enhances user interactivity by enabling dynamic features such as search filters, form validation, and real-time content updates.

3.1.2.1.2 Back-End Tools

The back-end ensures the logical and functional operation of the system.

- **PHP:** Manages data flow between the server and client, processes form submissions, and interacts with the database for user and job management.
- **MySQL:** Serves as the primary database system, storing user profiles, job listings, applications, and company details.

3.1.2.1.3 AI Integration

The platform incorporates AI functionalities to enhance recruitment accuracy and efficiency.

- **LM studio:** Used as the primary environment for AI model Deployment and integration.
- **Zephyr 7B Beta Model:** Powers the intelligent matching system by calculating compatibility percentages between job descriptions and candidate profiles, enabling automated applicant ranking.

3.1.2.1.4 Operation Platforms

- **Browser:** Microsoft Edge, Google Chrome, Mozilla Firefox
- **Operating Systems:** Windows 8, 10, 11; macOS

3.1.2.2 Hardware Requirements

- **RAM:** Minimum of 4 GB
- **Storage:** At least 100 MB of available disk space
- **Network:** Basic internet connectivity for server interaction and live functionality

3.1.3 Structuring System Requirements

3.1.3.1 Data Modeling

Data modeling helps visualize the internal structure of the database and its relationships. For JobConnect, an Entity-Relationship (ER) Diagram was developed to represent the major entities such as Users, Jobs, Employers, and Applications.

Each entity holds several attributes and is connected via relationships like one-to-many (e.g., one employer can post many jobs). The data model ensures that the information flow between the modules is efficient, secure, and scalable.

3.1.3.1.1 ER Diagram

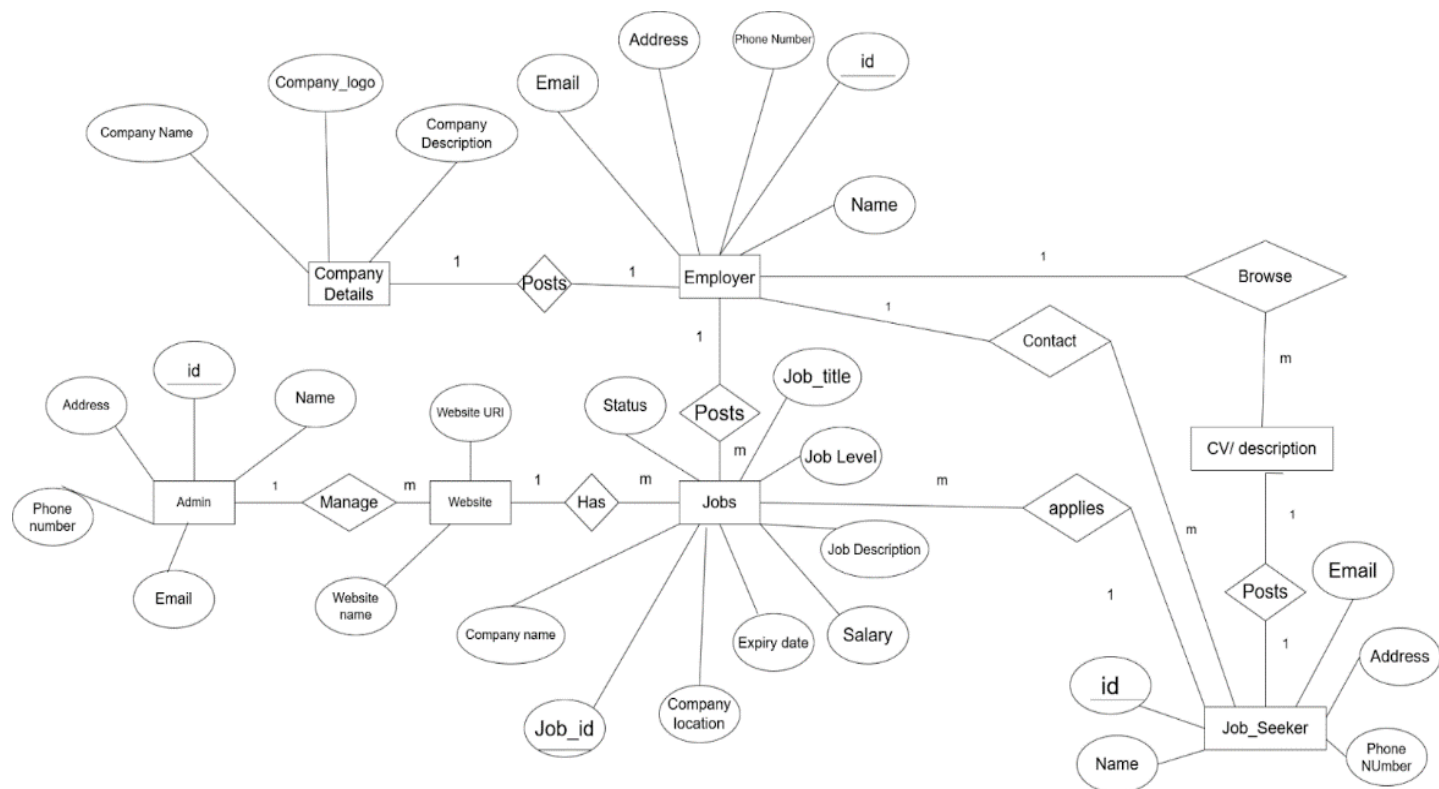


Figure 3-1 ER-Diagram

3.2 System Design

3.2.1 Designing system

3.2.1.1 Database Schema Design

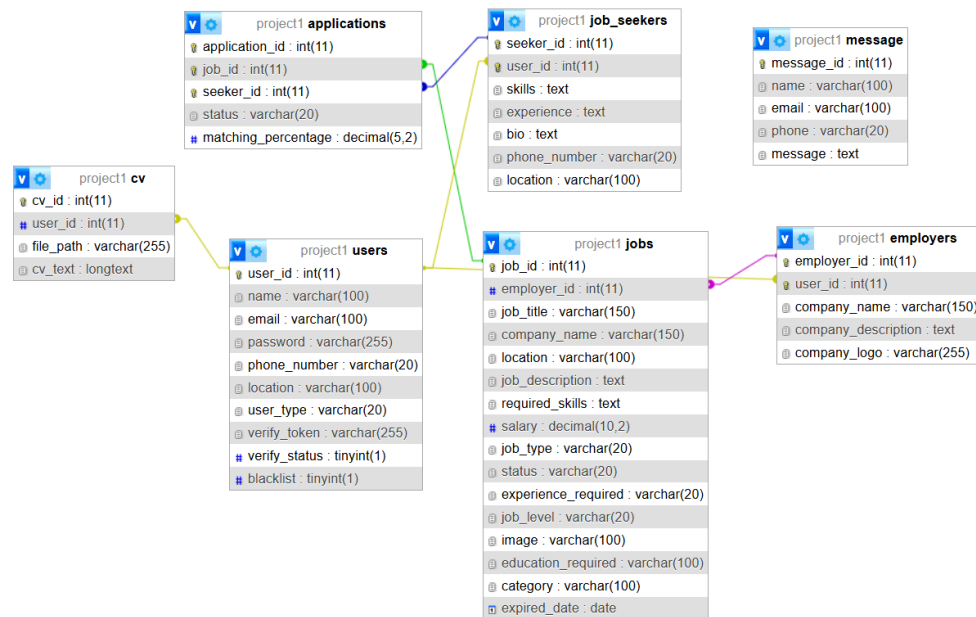


Figure 3-2 Database schema

3.2.1.2 Activity Diagram

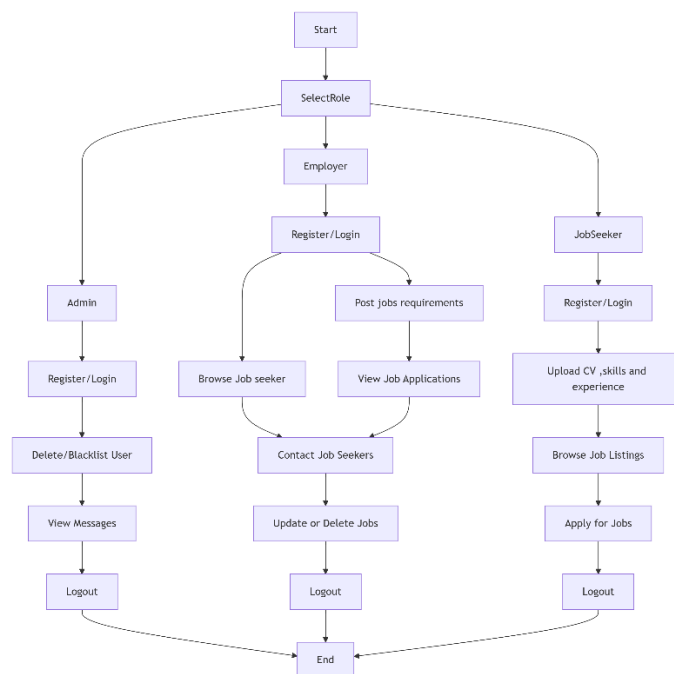


Figure 3-3 Activity Diagram

3.2.1.3 User-Case Diagram



Figure 3-4 Use-Case Diagram

3.2.1.4 Sequence-Diagram

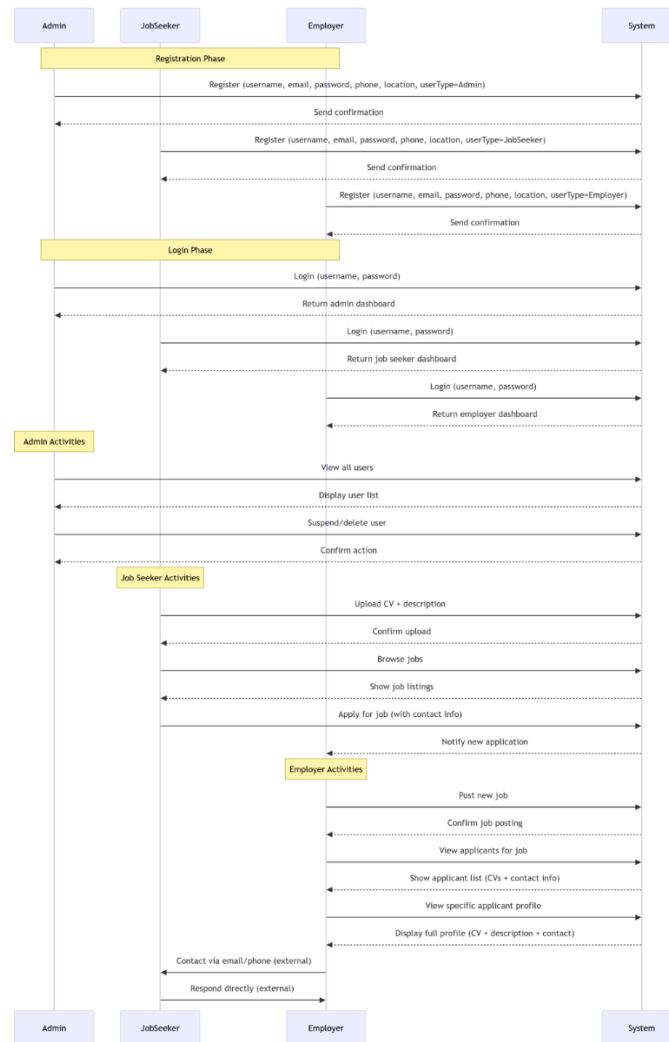


Figure 3-5 Sequence Diagram

3.2.2 Website Representation

3.2.2.1 Website design

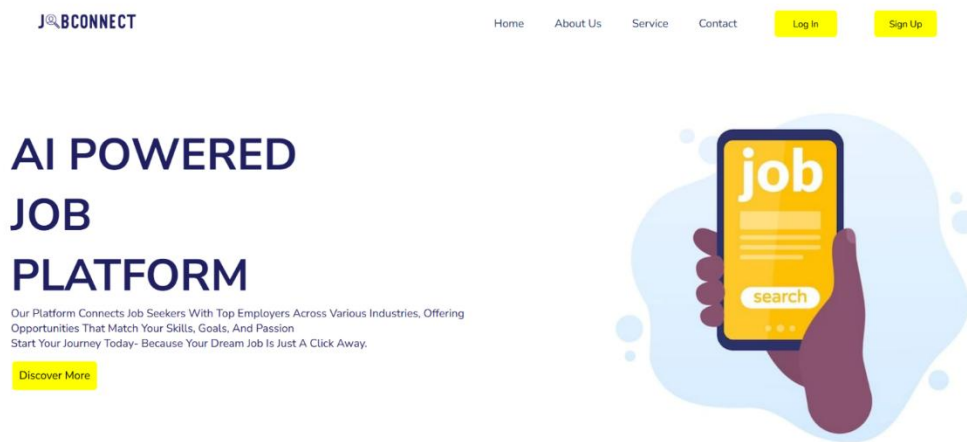


Figure 3-6 Website Design

3.2.2.2 Mobile Mockup

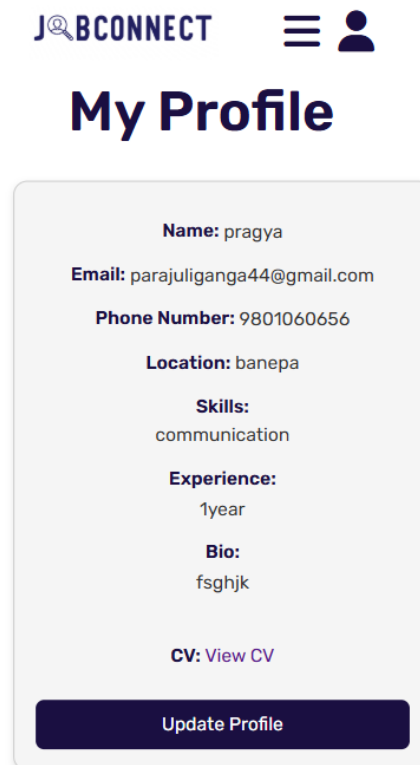


Figure 3-7 Job seeker page

JQBCONNECT  

ENTER COMPANY INFORMATION

Company Name

Company Description

Company Logo

No file chosen

Figure 3-8 Employer Page

3.2.3 System Architecture:

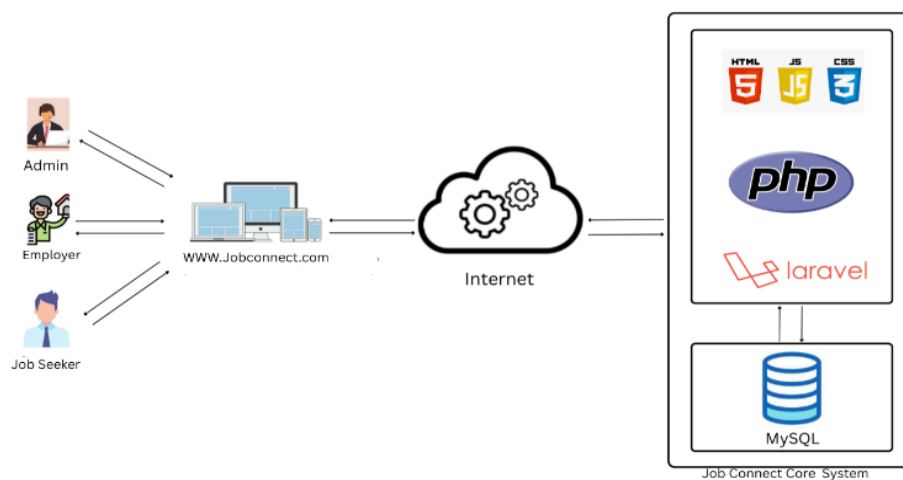


Figure 3-9 System Architecture

4 Chapter 4: Methodology

The development of JobConnect follows a structured methodology to deliver an efficient, user-friendly, and AI-powered job portal that effectively connects job seekers and employers. The key steps in the methodology include:

1. Literature Review:

Conducted a comprehensive review of existing job portals and recruitment platforms to understand their features, advantages, and limitations. This helped in identifying user needs and gaps, especially for localized and AI-enhanced job matching.

2. System Design:

Planned the overall architecture and workflow of JobConnect with a modular approach to ensure scalability and maintainability. Key design components included:

- User Roles and Permissions: Defined distinct functionalities for job seekers, employers, and administrators.
- Database Design: Developed schemas to manage users, job listings, applications, and communication data.
- User Interface: Designed intuitive, accessible pages for profile creation, job posting, search, and communication.
- AI Matching Module: Designed the integration layer for AI-powered candidate-job matching using LM Studio and the Zephyr 7B Beta model.

3. Technology Integration:

Selected and integrated technologies for seamless operation:

- Frontend: Built using HTML, CSS, and JavaScript for dynamic and responsive UI.
- Backend: Implemented with PHP to manage server-side logic and database interactions.
- Database: MySQL for reliable data storage and retrieval.
- AI Integration: Incorporated LM Studio running the Zephyr 7B Beta model to calculate match percentages and rank applicants.

4. Development:

Followed an incremental development model, implementing features in stages:

- User registration, authentication, and profile management.
- Job posting, searching, and application handling.
- AI matching and ranking functionality integration.
- Communication tools between employers and candidates.

5. Testing and Evaluation:

Conducted thorough system testing, including functional, usability, performance, and AI matching tests. User feedback was collected to identify areas for improvement, ensuring reliability and a positive user experience.

6. Deployment and Maintenance:

Deployed the platform on a web server with plans for ongoing maintenance, including performance monitoring, AI model updates, and future feature enhancements based on user needs and technological advancements

5 Chapter 5: System Testing

System testing is a crucial phase in web development that focuses on identifying discrepancies between expected outcomes and actual results when specific inputs are applied. This phase ensures the smooth functionality and features of the entire website. The primary goal is to evaluate the quality of the product and verify that it aligns with the intended design and user requirements.

During development, system testing involves applying various test scenarios to the fully integrated platform. For JobConnect, this testing validated the integrity of all components working together, ensuring seamless interaction, accuracy, and optimal performance. By focusing on system testing, we rigorously assessed the website's behavior, features, and performance to deliver a reliable and user-friendly job portal.

5.1 System Testing

System testing for JobConnect involved evaluating the fully integrated platform to ensure all features functioned as expected. Since the platform is web-based, testing was performed with active internet connectivity to reflect real-world usage. The aim was to ensure smooth and reliable operation under normal conditions, meeting both functionality and performance expectations.

S.N.	Test Case	Input/Action	Expected Output	Result
1	User Registration	New user fills signup form and submits	Successful account creation with email verification.	Pass
2	User login	Registered user enters valid credentials	Successful login redirection to dashboard	Pass
3	Profile Creation	User uploads CV, adds skills and personal details	Profile saved and visible under user account	Pass
4	Job Posting	Employer posts a new job with title, description and criteria	Job listing appears on platform and visible to job seeker	Pass
5	Job Search and Filtering	Job seeker searches for job with filters applied	Relevant jobs displaying matching search criteria	Pass
6	Application submission	Job seekers applies to a job	Application confirmation	Pass

			message, application stored in DB	
7	AI matching and Ranking	System runs matching algorithm between CV and job	Applicants are ranked based on compatibility percentage	Pass
8	Messaging between users	Employer sends email to job seeker	Email sent	Pass
9	Logout	User click logout	User session ends and redirect to homepage	Pass

Table 1: System Testing

5.2 Activity Matrix

Week	Activity Planned	Activity Achieved	Deviation and Causes
01 May-07 May	Discuss more about website we are creating and divided work	Started Frontend and design.	Internal Exams.
08 May-14 May	To develop necessary database.	Developed database table.	
15-21 May	To implement the email-verification	Developed registration and login functionality with email verification	
22-28 May	Develop the employer and jobseeker landing pages.	Developed the landing pages for employer and job-seeker.	
29 May-04 June	Enable CV re-upload, replacing the old file automatically	Job applications now require an uploaded CV. Users can update their CV anytime—new uploads overwrite the old file.	
05 June-11 June	Discuss to connect frontend and backend.	Connected the front-end and back-end.	
12 June- 18 June	To complete the function of each user.	Completed the function of each user.	
19 June-25 June	Added the expiry option for employer	Added the expiry option and if the job is expired jobseeker can't apply for job	
26 June- 02 July	Preparation of Progress report, and presentation.	Prepared Report and presentation.	

07July-19 July	Integrated AI using LM studio and model Zephyr 7B beta.	Matching percentage is calculated and user are sort accordingly.	
19 July-August 9			Board Exam preparation and board exam.
August 10-August 12	Preparation of Project Report and Presentation.	Completed the final report and presentation.	

Table 2 Activity Matrix

5.3 Gantt Chart

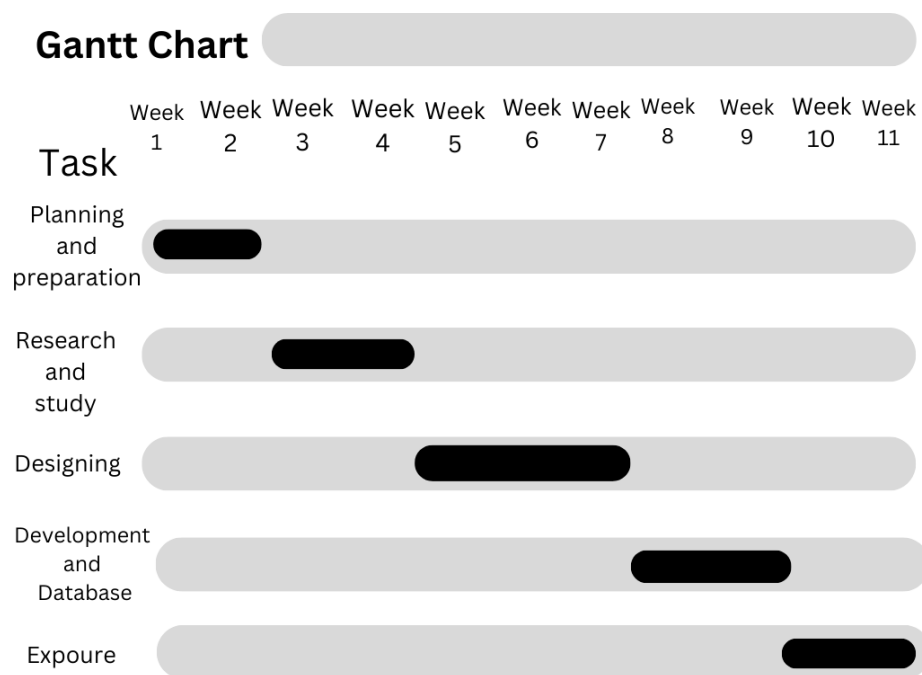


Figure 5-1 Gantt Chart

6 Chapter 6: Work done

6.1 Discussion on Achievements

6.1.1 Challenges:

As students developing the **JobConnect** platform, we encountered several technical and organizational difficulties throughout the process. Our limited experience with full-stack development presented a steep learning curve. Understanding the architecture of a multi-role system (Admin, Employer, Job Seeker) took time, as it required mastering backend integration, session control, and database design.

We also struggled with coordinating team tasks and schedules. Balancing academic workloads and project meetings often led to delays. Despite these obstacles, we continued to make progress by supporting one another and gradually improving our technical skills through testing, and teamwork.

6.1.2 Features:

This section outlines the major functionalities that have been successfully implemented in the "JobConnect" web platform. The website is designed to facilitate interaction between job seekers and employers, with administrative oversight for user and content management.

6.1.2.1 User Authentication and Role-Based Access

The platform supports a secure registration and login system for three user types: Admin, Employer, and Job Seeker. Each user type is authenticated and redirected to their respective dashboards upon successful login. New users can register, while existing users can securely access their accounts using login credentials.

If an Admin deletes a user, that user is automatically blacklisted. Blacklisted users are denied future login access, effectively ensuring platform security and integrity.

6.1.2.2 Admin Functionalities

Once logged in, the Admin is directed to an overview dashboard displaying critical system metrics, including:

- Total number of registered users
- Total employers
- Total job seekers
- Number of blacklisted users

The Admin can manage platform users through the "Manage Users" section. From here, they can delete users, which results in those users being blacklisted and barred from logging in again. Additionally, Admins can view user-submitted feedback, helping them monitor platform performance and user experience.

6.1.2.3 Employer Functionalities

Employers are directed to the Employer Page after login. Here, they are required to submit essential company details, including:

- Company Name
- Company Description
- Company Logo

After submitting company information, employers gain access to job posting features. They can:

- Create job postings with specific requirements
- Define job status and expiry date
- View, update, or delete the jobs they have posted

Employers can also view a list of applicants who applied for specific jobs. For each applicant, they can:

- View the CV
- Update application status (e.g., selected, rejected)
- Contact applicants via email directly from the platform

Additionally, employers can browse through job seekers who have not applied to any of their job postings. By reviewing their CVs, employers can proactively reach out to suitable candidates.

6.1.2.4 Job Seeker Functionalities

Job seekers, upon login, are taken to their personal dashboard. Before applying for any jobs, they are required to upload their CV. Once uploaded, job seekers can:

- Browse available job listings
- Apply to active (non-expired) job posts
- View company logos for a branded experience
- Update their personal profile information

Furthermore, job seekers can track all the jobs they have applied for under the “My Applications” section.

7 Chapter 7: Works to be done

7.1 Limitation:

While developing *JobConnect*, we focused on building a functional platform that connects employers with job seekers. However, due to time constraints and our current level of experience, there are a few areas where the system still has limitations:

- **No Instant Notifications**
Right now, users don't receive real-time notifications. For example, if a job seeker gets selected or if a new job is posted, they have to log in and check manually. This can cause delays in communication and missed opportunities.
- **Expired Job Posts Stay Visible**
When a job expires, job seekers can no longer apply, which is good. However, employers aren't reminded or prompted to update or remove those posts, so some old listings might remain unless they're manually handled.
- **No Built-in Messaging**
Employers can contact job seekers via email, but there's no messaging system inside the platform itself. As a result, communication takes place outside the site and isn't tracked within the system.
- **Additional Limitations:**
Other functionalities and performance-related exist due to project scope and resource constraints, such as optimization of AI matching speed.

While these limitations highlight areas where our website falls short, they also present opportunities for future improvements. Addressing these gaps would greatly enhance the both users experience and increase the potential for this platform to succeed in the competitive market.

7.2 Future Work:

This project opens the door for exciting improvements in future versions of *JobConnect*. These additions will make the platform smarter, more efficient, and easier to use for all types of user

- **Real-Time Notifications**
In future versions, we aim to implement real-time alerts for key activities—such as application updates, new job posts, or status changes. This would keep users informed without having to constantly check their dashboard.
- **In-Platform Messaging System**
Currently, communication between employers and job seekers happens through external email. We plan to introduce a built-in messaging system so that users can easily connect, ask questions, and follow up—all from within the platform.
- **Optimized AI Matching Performance:**

The current AI matching process can take considerable time to calculate compatibility percentages. Future work will focus on improving the speed and scalability of this system through optimized algorithms, more efficient data handling, and possibly distributed computing solutions. Faster Matching will enhance the overall user experience during peak usage.

Incorporating these improvements would make the platform more user-friendly, efficient, and adaptable, giving it the potential to succeed in the competitive online job portal space.

8 Chapter 8: Revenue Model

The sustainability and growth of JobConnect rely on a strategic revenue model designed to balance user accessibility with business viability. The platform adopts a diversified approach to generate income while providing value to both employers and job seekers.

1. Subscription Plans for Employers:

JobConnect offers tiered subscription packages tailored to employers' recruitment needs:

- **Basic Plan:** Allows a limited number of job postings with standard visibility, suitable for small businesses.
- **Premium Plan:** Provides increased posting limits, featured job listings, access to advanced candidate matching, and priority customer support.
- **Enterprise Plan:** Customized solutions for large organizations with unlimited postings, detailed analytics, and dedicated account management.

2. Featured Listings and Promotions:

Employers can opt to pay additional fees to promote their job listings. Sponsored posts receive higher visibility by appearing prominently on the platform and within search results, helping attract more qualified candidates efficiently.

3. Advertising Revenue:

JobConnect generates additional income by hosting relevant advertisements from related service providers such as training institutes, certification programs, and career development services, targeting both job seekers and employers.

This revenue model ensures a steady income stream while maintaining an affordable and effective platform for all users, enabling ongoing enhancements and expansion.

9 Chapter 9: Conclusions

9.1 Conclusions

Developing JobConnect has been a valuable and rewarding journey that helped us understand both the technical and human aspects of building a job portal tailored for local communities. Despite facing challenges with technology integration and teamwork coordination, we successfully created a functional platform that bridges the gap between employers and job seekers, especially at the grassroots level.

Our system supports multiple user roles with secure authentication, streamlined job posting, and application management features, addressing key pain points experienced by small businesses and individuals seeking employment. While some features like instant notifications and AI-based CV screening remain to be developed, the foundation we built offers a strong starting point for future improvements.

JobConnect focus on local employment and skill-based matching makes it stand out from existing platforms, promoting economic growth and inclusivity within communities. We believe that with continued enhancements and user engagement, this platform has the potential to significantly simplify the hiring process and empower users to find meaningful job connections.

Overall, this project has strengthened our technical skills, deepened our understanding of user-centered design, and highlighted the importance of collaboration in software development. We look forward to further refining JobConnect and contributing to the evolving landscape of digital employment solutions.

9.2 Recommendations

Developing a job portal that fully meets all user requirements is an ongoing challenge, as user needs and market trends continuously evolve. To ensure JobConnect remains relevant, effective, and user-friendly, the following key areas should be considered for future enhancements:

- **Technology Upgrades:** The platform should be regularly updated to incorporate emerging technologies. This includes adopting faster and more efficient frameworks, improving system performance, and enhancing scalability to support a growing user base. Staying aligned with technological advancements will help maintain competitiveness and reliability.
- **Enhanced Security:** With increasing cybersecurity threats, JobConnect must continually strengthen its security measures. Implementing features such as two-factor authentication (2FA), advanced data encryption, and secure communication channels will protect user data and build trust. Regular security audits and timely updates will safeguard the platform against potential vulnerabilities.

- **User Experience and Accessibility Improvements:** Enhancing the user interface (UI) and user experience (UX) by adding multilingual support and optimizing navigation, job search filters, and application processes will make the platform more accessible and intuitive. These improvements will help users find relevant opportunities and candidates more efficiently.
- **Expanded Features:** Introducing new functionalities such as real-time notifications, interview scheduling tools, employer and candidate rating systems, and multiple communication options will boost user engagement and convenience. These additions will also broaden the platform's appeal to diverse user groups, from freelancers to large enterprises.

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10 Appendix

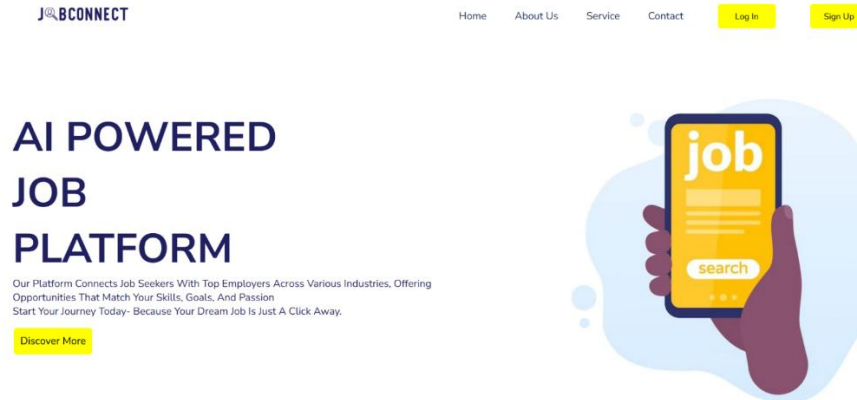


Figure 10-1 Home page

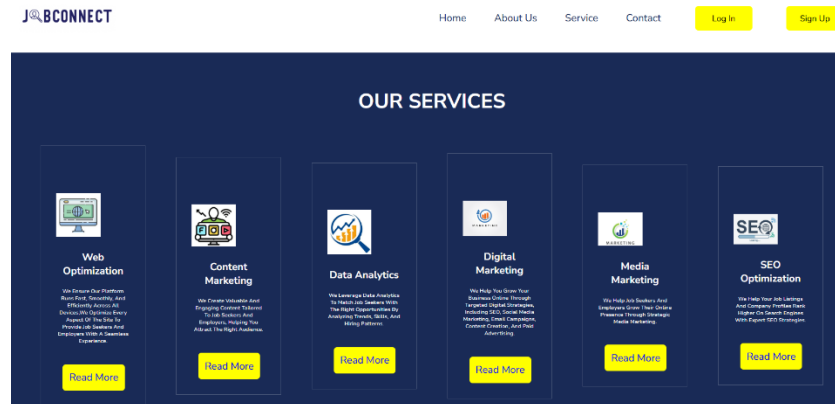


Figure 10-2 Our services



Figure 10-3 About Us page

Sign-Up

Username

Email

Password

Phone Number

Location

User Type

Sign Up*Figure 10-4 registration from***Login**

Username

Password

Log In[Don't Have A Account? Signup Now](#)*Figure 10-5 Login Form*

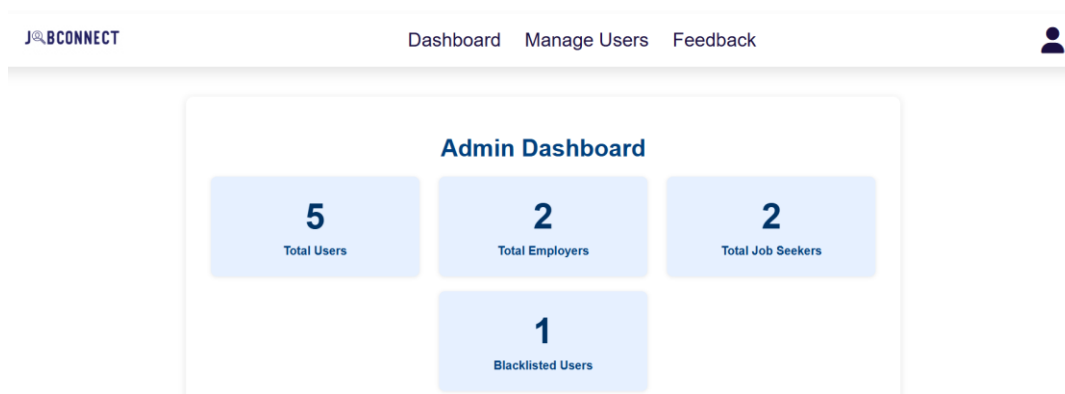


Figure 10-6 Admin Dashboard

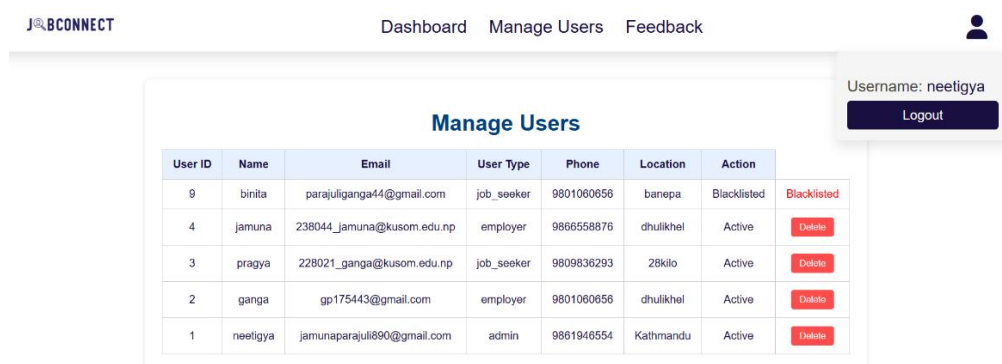


Figure 10-7 User managing Page

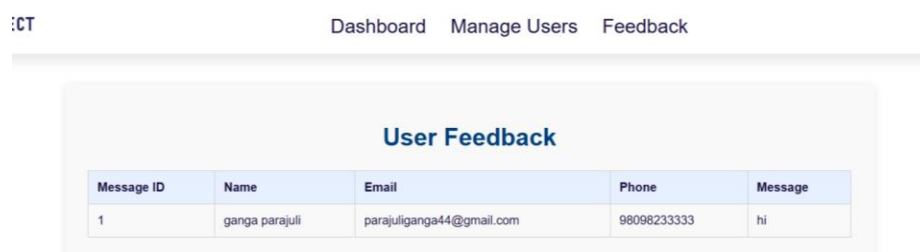


Figure 10-8 Admin Viewing feedback

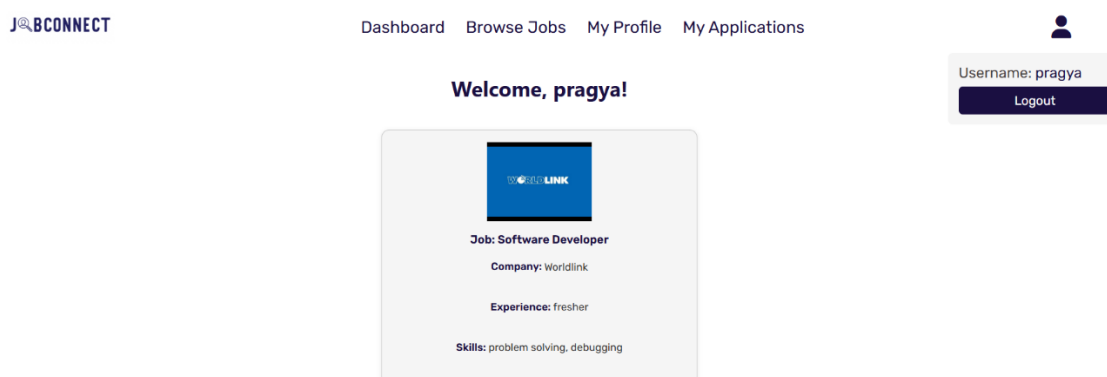


Figure 10-9 Job seeker dashboard



All Available Jobs by Category

 All Categories ▼

Category: general



Software Developer

Company: Worldlink

Location: Headoffice

Salary: 3000.00

Job Type: full-time

Job Level: entry

Experience Required: fresher

Education Required: Bachelor's degree in IT

Required Skills: problem solving, debugging

Expiry Date: August 25, 2025

Figure 10-10 Available Jobs



My Profile

Name: pragya

Email: 228021_ganga@kusom.edu.np

Phone Number: 9801060656

Location: dhulikhel

Skills:
communication, teamwork

Experience:
2year

Bio:
I'm a software developer with 2years of experience turning messy numbers into clear insights. I specialize in creating dashboards that help teams make better decisions faster. I'm currently looking for new opportunities where I can grow with a mission-driven company.

CV: [View CV](#)

[Update Profile](#)

Figure 10-11 Profile of Job seeker

Jobs You Have Applied To



Software Developer

Company: Worldlink

Location: Headoffice

Pending

Figure 10-12 Applied Jobs



ENTER COMPANY INFORMATION

Company Name

Company Description

Company Logo

Choose File No file chosen

Submit And Continue

Figure 10-13 Employer Dashboard



Job title

Company Name

Location

Required skills

Salary

Qualification required

Select Job type

Experience Required

Job Level

Username : ganga

Logout

Figure 10-14 Job adding Form



POSTED JOBS



Worldlink

Job Title: Software Developer

Location: Headoffice

Required skills: problem solving, debugging

Qualification Required: Bachelor's degree in IT

Job Type: full-time

Salary: Rs3000.00

Experience Required: fresher

Job Level: entry

Status: active

Expired Date: August 25, 2025

Job Description:

- Collaborate with other team members to develop high-quality software solutions.
- Write clean, efficient, and maintainable code.
- Stay up to date with emerging trends and technologies in web development.

Update

View Applicants

Delete

Figure 10-15 Posted Jobs

J@BCONNECT [Home](#) [Add Job](#) [My Jobs](#) [Job Seeker](#)

APPLICANTS FOR: SOFTWARE DEVELOPER

[Back to My Jobs](#)

Name	Email	Skills	Experience	Bio	Status	Matching %	CV	Update Status	Contact	Delete
pragya	228021_ganga@kusom.edu.np	communication, teamwork	2year	I'm a software developer with 2years of experience turning messy numbers into clear insights. I specialize in creating dashboards that help teams make better decisions faster. I'm currently looking for new opportunities where I can grow with a mission-driven company.	pending	3%	View CV	Pending Update	Contact	Delete

Figure 10-16 Applicants

J@BCONNECT [Home](#) [Add Job](#) [My Jobs](#) [Job Seeker](#)

ALL JOB SEEKERS

Username: pragya
Email: 228021_ganga@kusom.edu.np
Phone: 9801060656
Location: dhulikhel
Skills: communication, teamwork
Experience: 2year
Bio: I'm a software developer with 2years of experience turning messy numbers into clear insights. I specialize in creating dashboards that help teams make better decisions faster. I'm currently looking for new opportunities where I can grow with a mission-driven company.

[View CV](#)

[Contact Me](#)

Figure 10-17 Available Job seeker